# TerraMind on SmallMinesDS: Final Report

Nov 1, 2026 · Asım Emre Arslan

## Summary

We compared a frozen, pretrained TerraMind v1 tiny encoder with the same encoder trained from scratch for mine segmentation on SmallMinesDS 2022, using 1 to 1,491 (all) training images, in two experiments, and set both against a plain UNet of the same size in a third. **Experiment 1** trained every run for a fixed 50 epochs under four conditions (each encoder with and without D4 augmentation, 196 runs): augmentation improved the frozen encoder in every seed (+0.02 to +0.04 IoU) but had no consistent effect on the trainable one, so only the augmented conditions were kept. **Experiment 2** replaced the fixed budget with early stopping on test IoU and recorded the cost of every run (98 runs). In both experiments the frozen pretrained encoder gives the higher IoU at every N ≥ 10 in every seed (all images: 0.745 vs 0.673 in Experiment 1, 0.747 vs 0.688 in Experiment 2), and with 100 images it exceeds what the trainable encoder reaches with all 1,491. Early stopping gives the same IoU as the fixed 50 epochs from N = 250 on and 0.01–0.03 higher IoU at N = 3–100. The frozen encoder is also cheaper to train: 2.4 vs 5.5 GFLOP per training sample, and with all images 0.172 vs 0.370 GPU hours until early stopping. Experiment 3 trained a plain UNet with the same number of parameters from scratch under the Experiment 2 protocol: it beat both TerraMind models at every N and in every seed (all images: 0.789; one image: 0.579 vs 0.402), at 6.8 times the FLOPs per training sample of the frozen encoder. Three further baselines did not change this: a UNet with half the parameters matched the full one (0.789 with all images), the frozen SSL4EO ResNet50 CNN was slightly ahead of TerraMind (0.753) and the frozen DOFA transformer behind it (0.716), and the 2022 AlphaEarth embeddings classified per pixel with XGBoost reached 0.744.

## Setup (all experiments)

| Item | Setting |
|---|---|
| Data | SmallMinesDS 2022, 128 × 128 px patches; official split: 1,491 training, 644 test |
| Inputs | Sentinel-2 L2A (10 bands) + Sentinel-1 RTC (VV, VH, in dB); DEM not used. Experiment 4 ResNet50: S2 only; Experiment 5: 2022 AlphaEarth embeddings, 64 values per 10 m pixel |

| Item | Setting |
|---|---|
| Model | Experiments 1–2: TerraMind v1 tiny encoder (5.9 M parameters) + shallow UNet decoder (1.0 M), built with TerraTorch. Experiment 3: plain UNet (6.8 M); 3 v2: plain UNet (3.4 M); 4: DOFA base or SSL4EO-S12 ResNet50 encoder + the same decoder; 5: AlphaEarth embeddings + XGBoost |
| Encoder conditions | Pretrained, frozen (only the decoder learns) / random init, trainable; Experiments 3 and 3 v2: random init, trainable; Experiment 4: pretrained, frozen |
| Augmentation | D4: random rotation by 0/90/180/270°, with or without mirroring |
| Training images N | 1, 3, 5, 10, 20, 50, 100, 250 (5 seeds); 500, 1000, all = 1,491 (3 seeds) |
| Optimiser | AdamW, weight decay 0.01, batch size min(8, N) |
| Metric | Mine-class IoU on the 644 test patches; mean ± std over seeds |
| Hardware | NVIDIA GeForce RTX 3080 Laptop GPU (8 GB) |

Within a seed, the conditions use the same training images, decoder initialisation and batch order, so they are directly comparable. Code, results and reproduction steps: [github.com/asimarslan11/SmallMinesDS-Experiment-Project](github.com/asimarslan11/SmallMinesDS-Experiment-Project).

## Experiment 1: fixed training budget, four conditions

Every run trained for exactly 50 epochs with a cosine learning rate from 5e-3 to 1e-5, and the final model was evaluated; 4 conditions × 49 (seed, N) pairs = 196 runs, 5.0 GPU hours in total. The frozen pretrained encoder is ahead at every N ⩾ 10 in every seed, and augmentation adds +0.02 to +0.04 IoU to it at every N ⩾ 3 (5/5 seeds), while for the trainable encoder the effect changes sign with N. With 10 images, the frozen encoder with augmentation (0.591) matches the trainable encoder with 250 images (0.584–0.605).

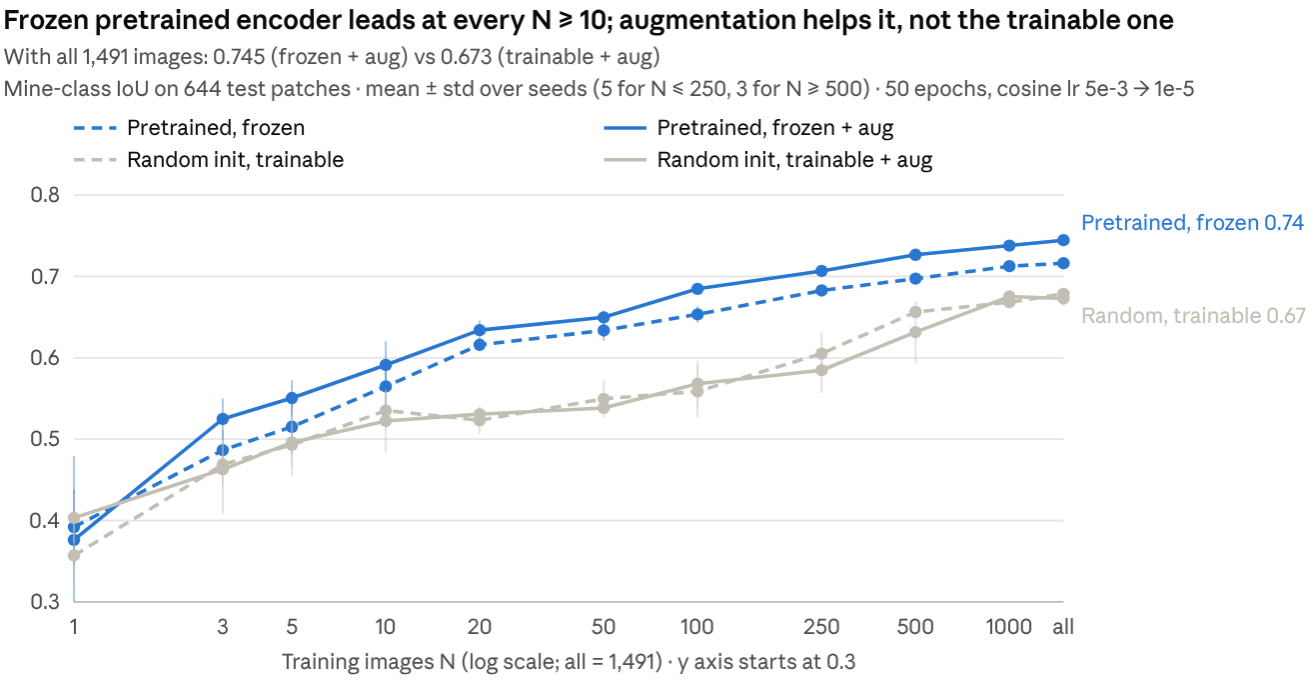



Figure 1. Experiment 1: mine-class IoU on the test set against the number of training images, mean ± std over seeds

**Table 1. Experiment 1 — mine-class IoU on the test set** (mean ± std over seeds).

| N | seeds | Pretrained, frozen | Pretrained, frozen + aug | Random init, trainable | Random init, trainable + aug |
|---|---|---|---|---|---|
| 1 | 5 | 0.392 ± 0.046 | 0.376 ± 0.103 | 0.357 ± 0.043 | 0.403 ± 0.030 |
| 3 | 5 | 0.487 ± 0.024 | 0.525 ± 0.025 | 0.469 ± 0.028 | 0.463 ± 0.054 |
| 5 | 5 | 0.515 ± 0.034 | 0.551 ± 0.022 | 0.493 ± 0.026 | 0.496 ± 0.040 |
| 10 | 5 | 0.565 ± 0.023 | 0.591 ± 0.029 | 0.535 ± 0.023 | 0.522 ± 0.039 |
| 20 | 5 | 0.616 ± 0.007 | 0.634 ± 0.012 | 0.523 ± 0.017 | 0.531 ± 0.007 |
| 50 | 5 | 0.633 ± 0.013 | 0.650 ± 0.008 | 0.550 ± 0.023 | 0.538 ± 0.005 |
| 100 | 5 | 0.653 ± 0.010 | 0.685 ± 0.007 | 0.559 ± 0.032 | 0.568 ± 0.028 |
| 250 | 5 | 0.683 ± 0.004 | 0.707 ± 0.002 | 0.605 ± 0.027 | 0.584 ± 0.028 |
| 500 | 3 | 0.697 ± 0.003 | 0.726 ± 0.003 | 0.656 ± 0.003 | 0.632 ± 0.038 |
| 1000 | 3 | 0.712 ± 0.003 | 0.738 ± 0.002 | 0.668 ± 0.006 | 0.675 ± 0.002 |
| all (1,491) | 3 | 0.716 ± 0.003 | 0.745 ± 0.003 | 0.679 ± 0.004 | 0.673 ± 0.009 |

**Table 2. Experiment 1 — IoU gain from D4 augmentation** (augmented minus unaugmented run with the same seed and training images; mean ± std, in brackets the seeds that improved).

| N | Pretrained, frozen | Random init, trainable |
|---|---|---|
| 1 | −0.016 ± 0.073 (3/5) | +0.046 ± 0.044 (4/5) |
| 3 | +0.038 ± 0.016 (5/5) | −0.006 ± 0.045 (2/5) |
| 5 | +0.035 ± 0.027 (5/5) | +0.003 ± 0.034 (3/5) |
| 10 | +0.027 ± 0.010 (5/5) | −0.013 ± 0.024 (2/5) |
| 20 | +0.018 ± 0.007 (5/5) | +0.008 ± 0.017 (3/5) |
| 50 | +0.016 ± 0.005 (5/5) | −0.011 ± 0.024 (2/5) |
| 100 | +0.031 ± 0.007 (5/5) | +0.010 ± 0.007 (5/5) |
| 250 | +0.024 ± 0.003 (5/5) | −0.021 ± 0.034 (2/5) |
| 500 | +0.029 ± 0.002 (3/3) | −0.025 ± 0.036 (1/3) |
| 1000 | +0.025 ± 0.002 (3/3) | +0.007 ± 0.008 (2/3) |
| all (1,491) | +0.029 ± 0.005 (3/3) | −0.006 ± 0.014 (2/3) |

## Experiment 2: early stopping and training cost

Only the two augmented conditions were run (98 runs, 4.1 GPU hours in total). Instead of a fixed budget, each run trained until its test IoU stopped improving: test IoU was checked every 5 epochs for every N; training stopped once 20 epochs (4 checks) passed without a gain above $10^{-3}$ over the best check; the learning rate started at 5e-3 and was halved at every non-improving check (minimum 1e-5); the model at the best check is reported. Steps, FLOPs and GPU hours were recorded for every run.

The frozen pretrained encoder is ahead at every N ⩾ 5 in every seed, 0.747 vs 0.688 with all images, and with 100 images (0.694) it already exceeds the trainable encoder with all 1,491. IoU rises with N for both encoders and flattens for the frozen one from N = 500 (0.729, 0.740, 0.747). Compared with Experiment 1 (Table 1, augmented columns), the frozen encoder reaches 0.01–0.03 higher IoU at N = 3–100 and the same IoU (within 0.005) from N = 250 on; the trainable encoder is equal or up to 0.03 higher. The frozen encoder costs 2.4 GFLOP per training sample against 5.5 for the trainable encoder, and with all images it stops after 0.172 GPU hours (284 TFLOP) against 0.370 (837 TFLOP).

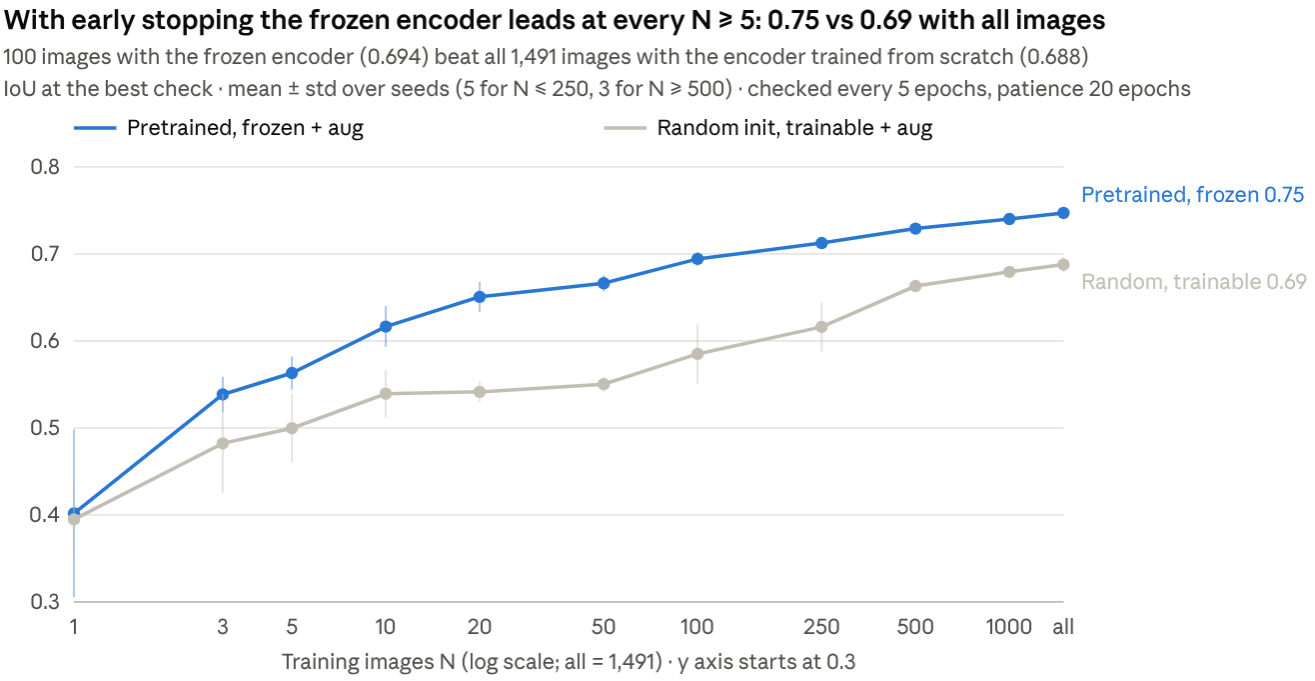



Figure 2. Experiment 2: mine-class IoU (best check) against the number of training images, mean ± std over seeds

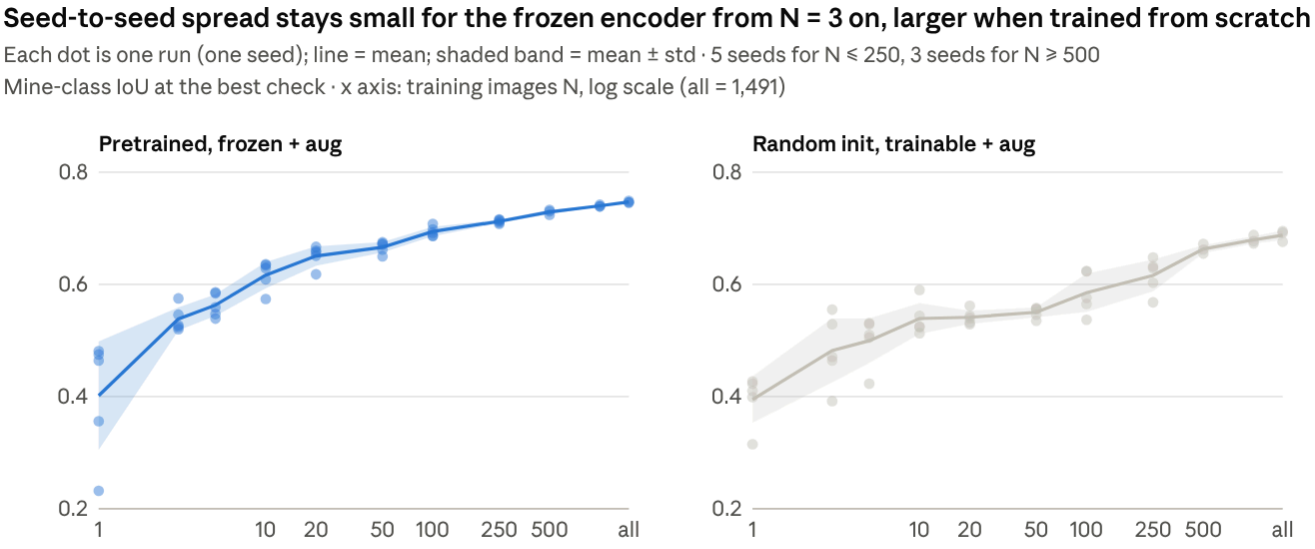



Figure 3. Experiment 2: seed-to-seed variance of IoU — every run as a dot, mean as a line, mean ± std as a shaded band

**Table 3. Experiment 2 — mine-class IoU on the test set** (best check; mean ± std over seeds).

| N | seeds | Pretrained, frozen + aug | Random init, trainable + aug |
|---|---|---|---|
| 1 | 5 | 0.402 ± 0.096 | 0.395 ± 0.041 |
| 3 | 5 | 0.539 ± 0.020 | 0.482 ± 0.057 |

| N | seeds | Pretrained, frozen + aug | Random init, trainable + aug |
|---|---|---|---|
| 5 | 5 | 0.563 ± 0.019 | 0.500 ± 0.040 |
| 10 | 5 | 0.616 ± 0.023 | 0.539 ± 0.027 |
| 20 | 5 | 0.651 ± 0.017 | 0.541 ± 0.012 |
| 50 | 5 | 0.666 ± 0.009 | 0.550 ± 0.009 |
| 100 | 5 | 0.694 ± 0.008 | 0.585 ± 0.034 |
| 250 | 5 | 0.712 ± 0.003 | 0.616 ± 0.028 |
| 500 | 3 | 0.729 ± 0.004 | 0.663 ± 0.007 |
| 1000 | 3 | 0.740 ± 0.002 | 0.680 ± 0.006 |
| all (1,491) | 3 | 0.747 ± 0.001 | 0.688 ± 0.008 |

**Table 4. Experiment 2 — training cost until early stopping, pretrained, frozen + aug** (mean ± std over seeds).

| N | best epoch | stop epoch | steps | GPU hours | total TFLOP |
|---|---|---|---|---|---|
| 1 | 43.0 ± 28.0 | 63.0 ± 28.0 | 63 ± 28 | 0.003 ± 0.001 | 15.4 ± 6.8 |
| 3 | 39.0 ± 9.2 | 59.0 ± 9.2 | 59 ± 9 | 0.002 ± 0.000 | 14.7 ± 2.3 |
| 5 | 55.0 ± 8.9 | 75.0 ± 8.9 | 75 ± 9 | 0.003 ± 0.000 | 19.0 ± 2.3 |
| 10 | 63.0 ± 10.8 | 83.0 ± 10.8 | 166 ± 22 | 0.005 ± 0.001 | 22.0 ± 2.9 |
| 20 | 76.0 ± 13.9 | 96.0 ± 13.9 | 288 ± 42 | 0.006 ± 0.001 | 27.7 ± 4.0 |
| 50 | 65.0 ± 17.0 | 85.0 ± 17.0 | 595 ± 119 | 0.010 ± 0.002 | 30.6 ± 6.1 |
| 100 | 57.0 ± 6.8 | 77.0 ± 6.8 | 1001 ± 88 | 0.015 ± 0.001 | 36.9 ± 3.2 |
| 250 | 62.0 ± 14.4 | 82.0 ± 14.4 | 2624 ± 459 | 0.034 ± 0.006 | 68.4 ± 12.0 |
| 500 | 53.3 ± 15.5 | 73.3 ± 15.5 | 4620 ± 974 | 0.059 ± 0.012 | 104.7 ± 22.1 |
| 1000 | 53.3 ± 4.7 | 73.3 ± 4.7 | 9167 ± 589 | 0.116 ± 0.007 | 191.8 ± 12.3 |
| all (1,491) | 55.0 ± 7.1 | 75.0 ± 7.1 | 14025 ± 1322 | 0.172 ± 0.017 | 283.5 ± 26.7 |

**Table 5. Experiment 2 — training cost until early stopping, random init, trainable + aug** (mean ± std over seeds).

| N | best epoch | stop epoch | steps | GPU hours | total TFLOP |
|---|---|---|---|---|---|
| 1 | 34.0 ± 23.5 | 54.0 ± 23.5 | 54 ± 24 | 0.002 ± 0.001 | 13.3 ± 5.8 |
| 3 | 49.0 ± 28.5 | 69.0 ± 28.5 | 69 ± 29 | 0.003 ± 0.001 | 17.8 ± 7.4 |

| N | best epoch | stop epoch | steps | GPU hours | total TFLOP |
|---|---|---|---|---|---|
| 5 | 40.0 ± 11.4 | 60.0 ± 11.4 | 60 ± 11 | 0.003 ± 0.001 | 16.1 ± 3.1 |
| 10 | 25.0 ± 17.6 | 45.0 ± 17.6 | 90 ± 35 | 0.004 ± 0.002 | 13.4 ± 5.2 |
| 20 | 35.0 ± 27.9 | 55.0 ± 27.9 | 165 ± 84 | 0.005 ± 0.002 | 19.4 ± 9.8 |
| 50 | 39.0 ± 5.8 | 59.0 ± 5.8 | 413 ± 41 | 0.010 ± 0.001 | 30.6 ± 3.0 |
| 100 | 49.0 ± 26.3 | 69.0 ± 26.3 | 897 ± 342 | 0.021 ± 0.008 | 54.9 ± 21.0 |
| 250 | 52.0 ± 19.1 | 72.0 ± 19.1 | 2304 ± 612 | 0.047 ± 0.012 | 117.2 ± 31.1 |
| 500 | 73.3 ± 12.5 | 93.3 ± 12.5 | 5880 ± 786 | 0.122 ± 0.015 | 281.3 ± 37.6 |
| 1000 | 75.0 ± 25.5 | 95.0 ± 25.5 | 11875 ± 3187 | 0.247 ± 0.069 | 549.7 ± 147.5 |
| all (1,491) | 78.3 ± 8.5 | 98.3 ± 8.5 | 18388 ± 1589 | 0.370 ± 0.031 | 836.6 ± 72.3 |

GPU hours are the wall-clock time of each run (model build, training and every test check). FLOPs are counted with torch.utils.flop_counter per sample (2.37 GFLOP per training sample for the frozen encoder, which has no backward pass through the encoder; 5.54 GFLOP for the trainable one; 1.87 GFLOP per evaluated sample) and multiplied by the samples processed; the total includes every test check (644 images each).

## Experiment 3: plain UNet of the same size

The third model is a standard UNet with no TerraMind component, trained from scratch: 6.83 M parameters (encoder 4.15 M, decoder 2.68 M), within 1 % of the 6.90 M of the TerraMind models. Its encoder halves the 128 × 128 input four times down to an 8 × 8 bottleneck, the same grid as TerraMind's tokens (channels 30/60/120/240/480, two 3 × 3 convolutions with BatchNorm and ReLU per stage); the decoder upsamples with transposed convolutions and skip connections. Data, inputs, training images, seeds, augmentation and the early-stopping protocol are exactly those of Experiment 2 (49 runs, 1.9 GPU hours).

The UNet gives the highest IoU at every N and in every seed, ahead of both TerraMind models: 0.789 vs 0.747 (frozen) and 0.688 (trainable) with all images, and 0.579 vs 0.402 with a single image. With 10 images (0.687) it is level with the frozen TerraMind encoder at 100 images (0.694), and with 250 images (0.753) it exceeds the frozen encoder with all 1,491. The gain costs compute: 16.1 GFLOP per training sample against 2.4 for the frozen encoder, so with all images it uses 1,806 vs 284 TFLOP and 0.227 vs 0.172 GPU hours.

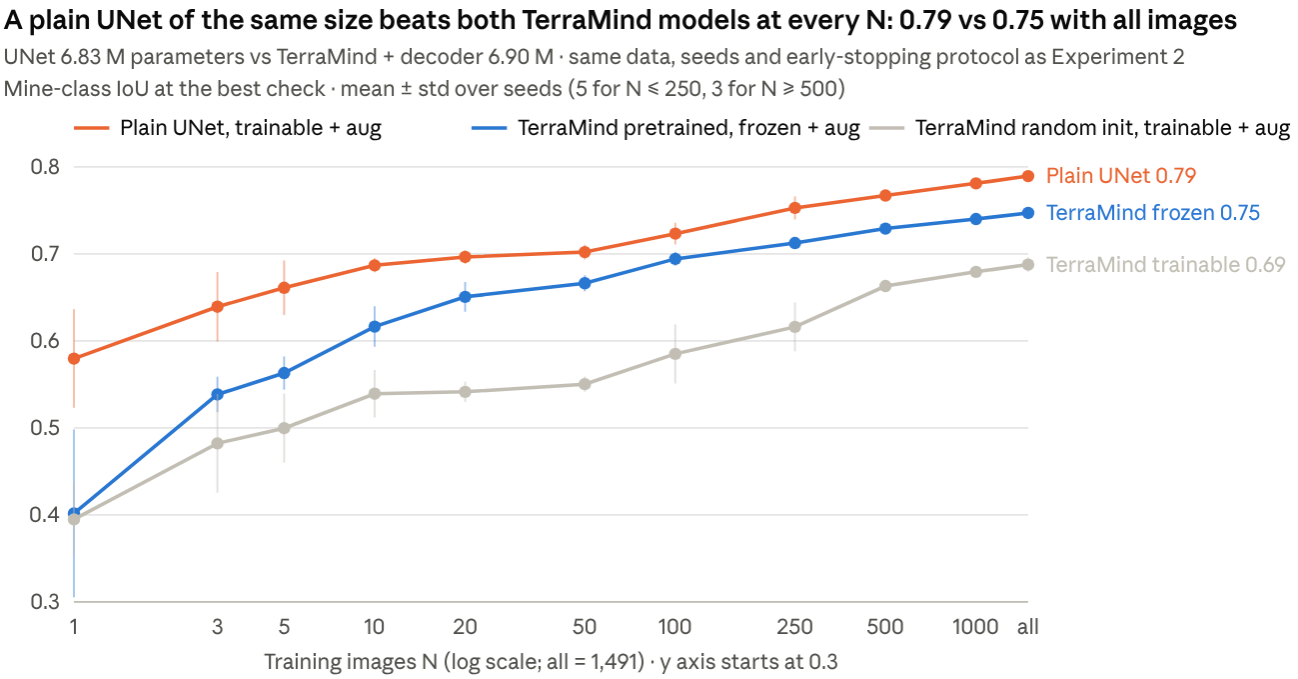



Figure 4. Experiment 3: mine-class IoU (best check) of the plain UNet against the two TerraMind models of Experiment 2, mean ± std over seeds

**Table 6. Experiment 3 — mine-class IoU on the test set** (best check; mean ± std over seeds; the TerraMind columns repeat Table 3).

| N | seeds | Plain UNet, trainable + aug | TerraMind pretrained, frozen + aug | TerraMind random init, trainable + aug |
|---|---|---|---|---|
| 1 | 5 | 0.579 ± 0.057 | 0.402 ± 0.096 | 0.395 ± 0.041 |
| 3 | 5 | 0.639 ± 0.040 | 0.539 ± 0.020 | 0.482 ± 0.057 |
| 5 | 5 | 0.661 ± 0.031 | 0.563 ± 0.019 | 0.500 ± 0.040 |
| 10 | 5 | 0.687 ± 0.008 | 0.616 ± 0.023 | 0.539 ± 0.027 |
| 20 | 5 | 0.696 ± 0.008 | 0.651 ± 0.017 | 0.541 ± 0.012 |
| 50 | 5 | 0.702 ± 0.009 | 0.666 ± 0.009 | 0.550 ± 0.009 |
| 100 | 5 | 0.723 ± 0.012 | 0.694 ± 0.008 | 0.585 ± 0.034 |
| 250 | 5 | 0.753 ± 0.013 | 0.712 ± 0.003 | 0.616 ± 0.028 |
| 500 | 3 | 0.767 ± 0.005 | 0.729 ± 0.004 | 0.663 ± 0.007 |
| 1000 | 3 | 0.781 ± 0.001 | 0.740 ± 0.002 | 0.680 ± 0.006 |
| all (1,491) | 3 | 0.789 ± 0.003 | 0.747 ± 0.001 | 0.688 ± 0.008 |

**Table 7. Experiment 3 — training cost until early stopping, plain UNet + aug** (mean ± std over seeds; 16.06 GFLOP per training sample, 5.39 GFLOP per evaluated sample).

| N | best epoch | stop epoch | steps | GPU hours | total TFLOP |
|---|---|---|---|---|---|
| 1 | 54.0 ± 7.3 | 74.0 ± 7.3 | 74 ± 7 | 0.003 ± 0.000 | 52.5 ± 5.2 |
| 3 | 56.0 ± 15.9 | 76.0 ± 15.9 | 76 ± 16 | 0.004 ± 0.001 | 56.4 ± 11.8 |
| 5 | 59.0 ± 30.7 | 79.0 ± 30.7 | 79 ± 31 | 0.004 ± 0.002 | 61.2 ± 23.8 |
| 10 | 46.0 ± 11.6 | 66.0 ± 11.6 | 132 ± 23 | 0.005 ± 0.001 | 56.4 ± 9.9 |
| 20 | 60.0 ± 15.2 | 80.0 ± 15.2 | 240 ± 45 | 0.007 ± 0.001 | 81.2 ± 15.4 |
| 50 | 59.0 ± 9.2 | 79.0 ± 9.2 | 553 ± 64 | 0.011 ± 0.001 | 118.2 ± 13.7 |
| 100 | 68.0 ± 18.9 | 88.0 ± 18.9 | 1144 ± 245 | 0.021 ± 0.005 | 202.3 ± 43.4 |
| 250 | 63.0 ± 12.9 | 83.0 ± 12.9 | 2656 ± 412 | 0.045 ± 0.007 | 390.7 ± 60.7 |
| 500 | 55.0 ± 7.1 | 75.0 ± 7.1 | 4725 ± 445 | 0.079 ± 0.008 | 654.1 ± 61.7 |
| 1000 | 55.0 ± 4.1 | 75.0 ± 4.1 | 9375 ± 510 | 0.150 ± 0.007 | 1256.2 ± 68.4 |
| all (1,491) | 53.3 ± 10.3 | 73.3 ± 10.3 | 13713 ± 1921 | 0.227 ± 0.042 | 1806.3 ± 253.1 |

## Experiments 3 v2, 4 and 5: further baselines

All three use the same training images, seeds and test set as Experiments 2 and 3; the deep-learning models also use the same augmentation and early-stopping protocol (49 runs per model).

- **Experiment 3 v2:** the plain UNet of Experiment 3 with half the parameters, 3.35 M (channels 21/42/84/168/336), trained from scratch.
- **Experiment 4:** two other frozen pretrained encoders in place of TerraMind, with the same UNet decoder: DOFA base, a 111 M-parameter vision transformer that adapts to each band by its wavelength (S2 + S1), and a ResNet50 CNN pretrained with MoCo on SSL4EO-S12 Sentinel-2 images (25.6 M; S2 only, as it has no S1 input). Their decoders are larger than TerraMind's (5.0 M and 4.2 M vs 1.0 M) because their features are wider.
- **Experiment 5:** the 2022 AlphaEarth Foundations embeddings (Google DeepMind; 64 values per 10 m pixel, read from the public Source Cooperative mirror) under every patch, classified per pixel with XGBoost (learning rate 0.1, depth 6). It is trained on the pixels of the same N images (at most 4.2 M pixels) and stopped on test IoU, checked every 25 rounds with a patience of 100 rounds.

None of the new models beats the plain UNet. The half-size UNet matches the full one from N = 10 on (0.789 with all images for both) at half the FLOPs per sample; only at $N \leqslant 5$ is it lower and more seed-dependent. Among the frozen encoders, the ResNet50 CNN is ahead of TerraMind in 42 of 49 runs, most clearly with few images (N = 10: 0.668 vs 0.616;

all images: 0.753 vs 0.747), while DOFA is the weakest pretrained model from N = 3 on (all images: 0.716). AlphaEarth + XGBoost is strong with very few images (0.551 with one image, second only to the plain UNet) but levels off at 0.74 from N = 500 on, level with TerraMind and the ResNet50.

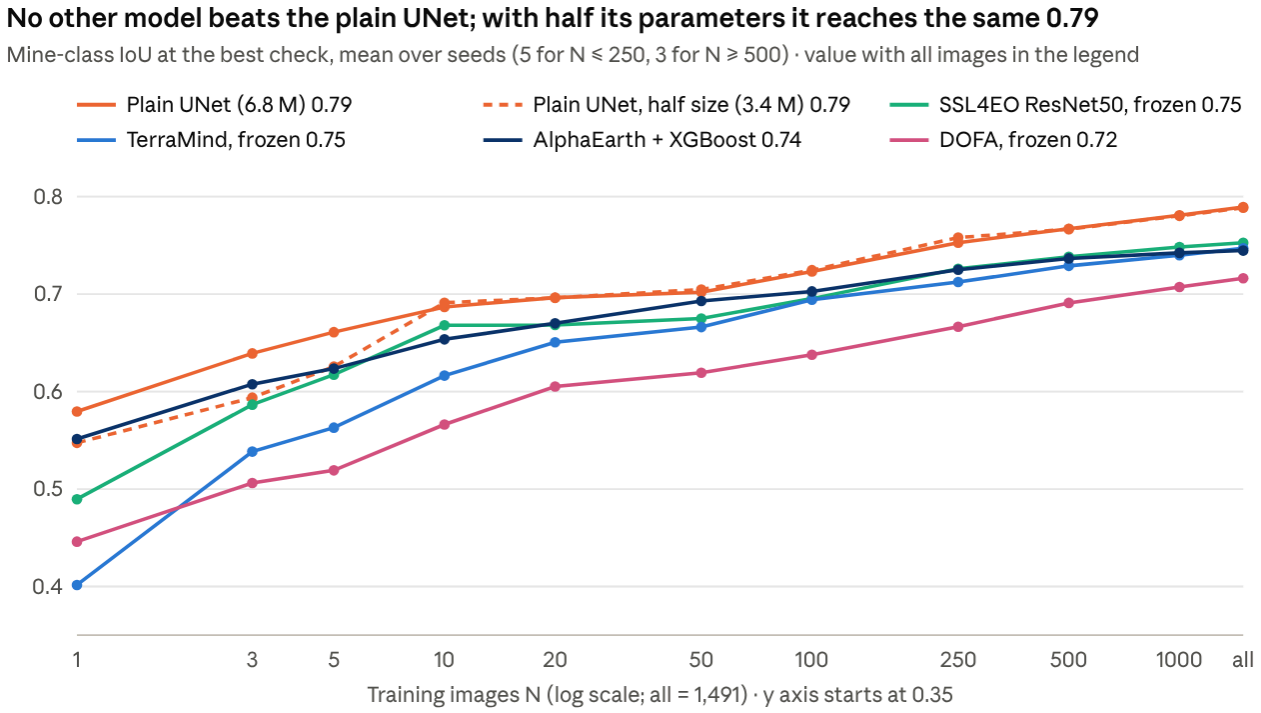



Figure 5. Experiments 3 v2, 4 and 5: mine-class IoU (best check) of every model against the number of training images, mean over seeds

**Table 8. Experiments 3 v2, 4 and 5 — mine-class IoU on the test set** (best check; mean ± std over seeds; columns ordered by IoU with all images; the plain UNet and TerraMind columns repeat Table 6).

| N | seeds | Plain UNet (C) | Plain UNet, half size (D) | SSL4EO ResNet50, frozen (F) | TerraMind, frozen (A) | AlphaEarth + XGBoost (G) | DOFA, frozen (E) |
|---|---|---|---|---|---|---|---|
| 1 | 5 | 0.579 ± 0.057 | 0.547 ± 0.095 | 0.490 ± 0.085 | 0.402 ± 0.096 | 0.551 ± 0.025 | 0.446 ± 0.043 |
| 3 | 5 | 0.639 ± 0.040 | 0.594 ± 0.080 | 0.587 ± 0.035 | 0.539 ± 0.020 | 0.608 ± 0.025 | 0.506 ± 0.021 |
| 5 | 5 | 0.661 ± 0.031 | 0.626 ± 0.027 | 0.617 ± 0.014 | 0.563 ± 0.019 | 0.624 ± 0.023 | 0.519 ± 0.022 |
| 10 | 5 | 0.687 ± 0.008 | 0.691 ± 0.009 | 0.668 ± 0.013 | 0.616 ± 0.023 | 0.653 ± 0.008 | 0.566 ± 0.018 |
| 20 | 5 | 0.696 ± 0.008 | 0.696 ± 0.003 | 0.669 ± 0.016 | 0.651 ± 0.017 | 0.670 ± 0.005 | 0.605 ± 0.006 |
| 50 | 5 | 0.702 ± 0.009 | 0.705 ± 0.016 | 0.675 ± 0.019 | 0.666 ± 0.009 | 0.693 ± 0.010 | 0.619 ± 0.004 |

| N | seeds | Plain UNet (C) | Plain UNet, half size (D) | SSL4EO ResNet50, frozen (F) | TerraMind, frozen (A) | AlphaEarth + XGBoost (G) | DOFA, frozen (E) |
|---|---|---|---|---|---|---|---|
| 100 | 5 | 0.723 ± 0.012 | 0.725 ± 0.013 | 0.695 ± 0.012 | 0.694 ± 0.008 | 0.703 ± 0.010 | 0.638 ± 0.002 |
| 250 | 5 | 0.753 ± 0.013 | 0.758 ± 0.005 | 0.726 ± 0.004 | 0.712 ± 0.003 | 0.725 ± 0.005 | 0.666 ± 0.004 |
| 500 | 3 | 0.767 ± 0.005 | 0.767 ± 0.007 | 0.739 ± 0.003 | 0.729 ± 0.004 | 0.737 ± 0.001 | 0.691 ± 0.001 |
| 1000 | 3 | 0.781 ± 0.001 | 0.780 ± 0.004 | 0.748 ± 0.002 | 0.740 ± 0.002 | 0.742 ± 0.001 | 0.708 ± 0.002 |
| all (1,491) | 3 | 0.789 ± 0.003 | 0.789 ± 0.004 | 0.753 ± 0.002 | 0.747 ± 0.001 | 0.744 ± 0.000 | 0.716 ± 0.001 |

**Table 9. Experiments 3 v2 and 4 — training cost until early stopping** (mean ± std over seeds; D = half-size UNet, E = DOFA, F = ResNet50). FLOPs per training / evaluated sample: D 7.9 / 2.7, E 14.9 / 13.3, F 5.1 / 3.8 GFLOP. These runs shared the computer with the XGBoost runs, so their GPU hours are somewhat inflated; the XGBoost runs took 2.9 h in total on 12 CPU threads.

| N | D: GPU hours | D: total TFLOP | E: GPU hours | E: total TFLOP | F: GPU hours | F: total TFLOP |
|---|---|---|---|---|---|---|
| 1 | 0.002 ± 0.001 | 18.6 ± 8.6 | 0.008 ± 0.003 | 138.6 ± 68.6 | 0.011 ± 0.003 | 39.1 ± 14.6 |
| 3 | 0.002 ± 0.001 | 22.4 ± 7.1 | 0.008 ± 0.002 | 130.4 ± 41.7 | 0.014 ± 0.001 | 53.8 ± 7.2 |
| 5 | 0.003 ± 0.001 | 29.5 ± 6.0 | 0.006 ± 0.001 | 94.0 ± 19.5 | 0.014 ± 0.002 | 51.4 ± 7.3 |
| 10 | 0.003 ± 0.001 | 28.3 ± 7.9 | 0.009 ± 0.001 | 139.9 ± 23.4 | 0.018 ± 0.004 | 55.5 ± 6.2 |
| 20 | 0.005 ± 0.001 | 37.6 ± 10.0 | 0.010 ± 0.002 | 153.1 ± 28.4 | 0.018 ± 0.004 | 46.2 ± 9.1 |
| 50 | 0.010 ± 0.002 | 71.6 ± 13.1 | 0.012 ± 0.002 | 156.8 ± 31.9 | 0.035 ± 0.005 | 65.7 ± 6.8 |
| 100 | 0.016 ± 0.003 | 94.1 ± 18.1 | 0.021 ± 0.005 | 235.0 ± 54.2 | 0.040 ± 0.007 | 63.6 ± 9.1 |
| 250 | 0.042 ± 0.007 | 199.6 ± 26.9 | 0.036 ± 0.006 | 357.8 ± 55.4 | 0.084 ± 0.014 | 107.2 ± 20.4 |
| 500 | 0.067 ± 0.006 | 293.8 ± 26.8 | 0.077 ± 0.014 | 713.9 ± 125.3 | 0.092 ± 0.026 | 166.8 ± 21.4 |
| 1000 | 0.142 ± 0.012 | 605.4 ± 51.5 | 0.155 ± 0.010 | 1374.1 ± 98.9 | 0.154 ± 0.033 | 381.5 ± 80.0 |
| all (1,491) | 0.271 ± 0.048 | 1112.9 ± 187.6 | 0.221 ± 0.022 | 1941.3 ± 186.5 | 0.204 ± 0.044 | 498.7 ± 106.1 |